



```
In [2]: l = []  
l
```

```
Out[2]: []
```

```
In [3]: len(l)
```

```
Out[3]: 0
```

```
In [4]: l1 = [10, 20, 30]  
l1
```

```
Out[4]: [10, 20, 30]
```

```
In [5]: len(l1)
```

```
Out[5]: 3
```

```
In [6]: l1
```

```
Out[6]: [10, 20, 30]
```

```
In [7]: l1.append(40)  
l1.append(50)  
l1.append(60)  
l1.append(10)
```

```
In [8]: l1
```

```
Out[8]: [10, 20, 30, 40, 50, 60, 10]
```

```
In [9]: len(l1)
```

```
Out[9]: 7
```

```
In [10]: l1.append(70, 80)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[10], line 1  
----> 1 l1.append(70, 80)  
  
TypeError: list.append() takes exactly one argument (2 given)
```

```
In [11]: l1
```

```
Out[11]: [10, 20, 30, 40, 50, 60, 10]
```

```
In [12]: l1.append(['a', 'b', 'c'])
```

```
In [13]: l1
```

Out[13]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

```
In [14]: l2 = [34, 5.3, 'jyothi', 1+2j, True]
         l2
```

Out[14]: [34, 5.3, 'jyothi', (1+2j), True]

```
In [15]: print(l)
         print(l1)
         print(l2)

[]
[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
[34, 5.3, 'jyothi', (1+2j), True]
```

```
In [16]: print(len(l))
         print(len(l1))
         print(len(l2))
```

0
8
5

```
In [17]: print(id(l))
         print(id(l1))
         print(id(l2))
```

2094342054016
2094342052672
2094342514304

```
In [18]: print(type(l))
         print(type(l1))
         print(type(l2))
```

<class 'list'>
<class 'list'>
<class 'list'>

```
In [19]: l1
```

Out[19]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]

```
In [20]: # list membership

10 in l1
```

Out[20]: True

```
In [21]: ['a', 'b', 'c'] in l1
```

Out[21]: True

```
In [22]: l1
```

```
Out[22]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [23]: 100 in l1
```

```
Out[23]: False
```

```
In [24]: l2
```

```
Out[24]: [34, 5.3, 'jyothi', (1+2j), True]
```

```
In [25]: l3 = l2.copy()
```

```
In [26]: l3
```

```
Out[26]: [34, 5.3, 'jyothi', (1+2j), True]
```

```
In [27]: print(id(l2))  
         print(id(l3))
```

```
2094342514304
```

```
2094342508352
```

```
In [28]: print(l)  
         print(l1)  
         print(l2)  
         print(l3)
```

```
[]
```

```
[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
[34, 5.3, 'jyothi', (1+2j), True]
```

```
[34, 5.3, 'jyothi', (1+2j), True]
```

```
In [29]: l3
```

```
Out[29]: [34, 5.3, 'jyothi', (1+2j), True]
```

```
In [30]: len(l3)
```

```
Out[30]: 5
```

```
In [31]: l3.clear()
```

```
In [32]: l3
```

```
Out[32]: []
```

```
In [33]: len(l3)
```

```
Out[33]: 0
```

```
In [34]: del l3
```

```
In [35]: print(l)
         print(l1)
         print(l2)
```

```
[]
[10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
[34, 5.3, 'jyothi', (1+2j), True]
```

```
In [36]: l2.count(5.3)
```

```
Out[36]: 1
```

```
In [37]: l1.count(10)
```

```
Out[37]: 2
```

```
In [38]: l1
```

```
Out[38]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [39]: l1
```

```
Out[39]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [40]: l1[:]
```

```
Out[40]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [41]: l1[-1]
```

```
Out[41]: ['a', 'b', 'c']
```

```
In [42]: l1
```

```
Out[42]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [43]: l1[100]
```

```
-----
IndexError                                Traceback (most recent call last)
Cell In[43], line 1
----> 1 l1[100]

IndexError: list index out of range
```

```
In [44]: l1[4:]
```

```
Out[44]: [50, 60, 10, ['a', 'b', 'c']]
```

```
In [45]: l1
```

```
Out[45]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [46]: l1[3:-1]
```

```
Out[46]: [40, 50, 60, 10]
```

```
In [47]: l1
```

```
Out[47]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [48]: l1[0:10:3]
```

```
Out[48]: [10, 40, 10]
```

```
In [49]: l1
```

```
Out[49]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [50]: l1
```

```
Out[50]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [51]: l1[::2]
```

```
Out[51]: [10, 30, 50, 10]
```

```
In [52]: l1
```

```
Out[52]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [53]: l1[::3]
```

```
Out[53]: [10, 40, 10]
```

```
In [54]: l1
```

```
Out[54]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [55]: l1.count(10)
```

```
Out[55]: 2
```

```
In [56]: l1
```

```
Out[56]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [57]: l1[::-1]
```

```
Out[57]: [['a', 'b', 'c'], 10, 60, 50, 40, 30, 20, 10]
```

```
In [58]: l1
```

```
Out[58]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [59]: l1[::-2]
```

```
Out[59]: [['a', 'b', 'c'], 60, 40, 20]
```

```
In [60]: l1
```

```
Out[60]: [10, 20, 30, 40, 50, 60, 10, ['a', 'b', 'c']]
```

```
In [61]: l1[::-3]
```

```
Out[61]: [['a', 'b', 'c'], 50, 20]
```

```
In [ ]:
```